# C-GEAR: Calibrated Genetic Efficiency-Aware Rank Allocation for Incremental LoRA Fine-Tuning

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**Abstract**

Incremental LoRA avoids fixing a large adapter rank in advance, but IncreLoRA assigns each new component through a local importance-based Greedy rule and continues growing toward a preset target. We introduce *C-GEAR*, which anchors evolutionary candidate generation at that rule, calibrates candidate updates on training data only, selects a budget-feasible variable event size (including no growth), and stops allocation when positive growth is repeatedly unreliable. On GLUE RTE with DeBERTa-v3-base and six matched seeds, C-GEAR improves mean selected-checkpoint accuracy from 87.30% to 88.09% while reducing selected active parameters by 3.55% on average per matched pair. Terminal allocation trajectories use 13.86% fewer active parameters. These descriptive results establish a favorable local accuracy–efficiency trade-off without claiming broad or statistically significant superiority.

## 1 Introduction

Low-rank adaptation (LoRA) freezes a pretrained model and learns low-rank updates, sharply reducing task-specific storage [3]. Yet a uniform fixed rank ignores that attention and feed-forward matrices need not contribute equally. IncreLoRA instead begins small and allocates rank during optimization [7], but its locally Greedy allocator asks only which modules currently look most important; it cannot compare coordinated alternatives or decline an allocation event.

This work asks whether incremental rank growth can become more selective and parameter-efficient without sacrificing downstream performance. C-GEAR answers with (i) Greedy-anchored evolutionary search over non-uniform module sets, (ii) conservative training-only calibration, (iii) adaptive event size $k$ including $k = 0$, and (iv) explicit budget and stopping logic. Beyond the allocator, the project contributes dynamic-checkpoint reconstruction, candidate snapshot/restore, budget-correct parameter accounting, regression tests, and a schema-validated telemetry/analysis pipeline. Code and reproducibility artifacts are available at `https://github.com/Aliflori/colabLoRA`.

## 2 Related Work

Adaptive LoRA methods relax a single hand-chosen rank in different ways. AdaLoRA allocates a global budget by importance [8]; DyLoRA jointly trains a nested range of usable ranks [5]; SoRA learns sparse rank gates [1]; AutoLoRA uses meta-learning to tune layer ranks [9]; and DoRA dynamically prunes structured rank-one components [4]. IncreLoRA reverses pruning by progressively adding components, avoiding a restrictive initial upper rank [7]. C-GEAR retains this incremental setting but targets a different gap: *which combination* should grow at an event, by how much, and whether growth is warranted at all.

## 3 Method / Proposed Approach

Let module $i$ have active rank $r_i$ and dimensions $d_{\text{in},i}$ and $d_{\text{out},i}$. Its active adaptation cost is

$$C(\mathbf{r}) = \sum_i r_i(d_{\text{in},i} + d_{\text{out},i}), \qquad C(\mathbf{r}) \le B. \tag{1}$$

At an allocation event, C-GEAR constructs legal module sets for $k \in \{0, \dots, K\}$. A repaired Greedy solution is retained as a reference, while greedy-neighborhood, cost-aware, diversity-aware, and population-search candidates explore non-Greedy combinations. Set crossover, add/remove/replace mutation, elitism, and Hamming-diversity pressure operate under hard projected-cost feasibility; no post-hoc local search is used.

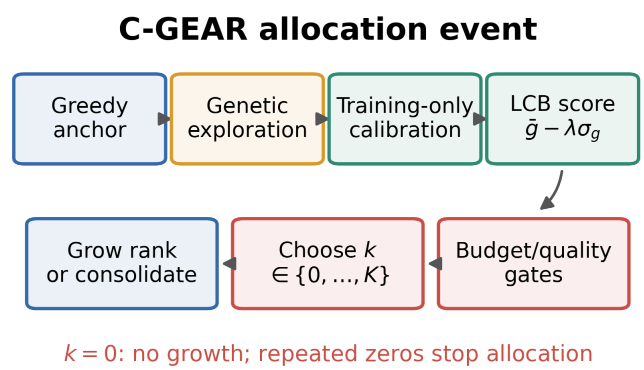


**Figure 1.** One C-GEAR allocation event. Candidate calibration never uses validation labels; repeated reliable $k = 0$ decisions trigger consolidation rather than later rank growth.

Structural search alone is not trusted to commit a change. Each shortlist candidate is virtually materialized, updated on three held-out *training* folds, and restored with model, optimizer, rank, and random-number state unchanged. For loss decrease $g_j$ on fold $j$, the implemented conservative score is

$$S_{\text{cal}} = \bar{g} - \lambda\sigma_g, \qquad \lambda = 0.5. \tag{2}$$

A positive candidate must clear reliability, Greedy-quality, marginal-gain, and budget gates. The quality set is then ordered by projected active cost before calibrated quality. If none qualifies, $k = 0$ leaves the rank pattern unchanged; two consecutive no-growth events stop allocation and begin ordinary consolidation training. Figure 1 summarizes the decision path.

## 4 Experimental Setup

We perform a controlled local comparison on GLUE RTE [6] with `microsoft/deberta-v3-base` [2]. Greedy In-

**Table 1.** Six-seed RTE results (mean ± sample SD). Accuracy and selected counts describe the same selected checkpoint; final counts are architectural.

| Metric | Greedy IncreLoRA | C-GEAR |
|---|---|---|
| Accuracy ↑ | 87.30 ± 0.84 | **88.09** ± 1.14 |
| Selected params ↓ | 828.0k ± 65.2k | **795.2k** ± 37.1k |
| Selected rank ↓ | 99.0 ± 27.8 | **86.7** ± 14.5 |
| Final params ↓ | 933.6k ± 5.3k | **804.1k** ± 39.3k |
| Final rank ↓ | 144.0 ± 0.0 | **89.7** ± 15.0 |

creLoRA and C-GEAR use identical seeds 41–46, data order, model, targets, and optimization: 25 epochs (1,950 steps), sequence length 192, train/evaluation batches 32/8, AdamW at $1.2\times10^{-3}$, linear scheduling with 100 warm-up steps, weight decay 0.01, and FP16 on an RTX 4060 Laptop GPU. Allocation begins from rank one in 72 target modules and is considered every 100 steps. C-GEAR uses population 12, four generations, calibration top-6, budget ratio 0.94, and the fixed settings recorded by telemetry. RTE validation accuracy is the performance metric.

Efficiency is *active model parameters*: the fixed task head plus components represented by the active rank pattern, not raw `requires_grad` storage. "Selected" denotes the checkpoint chosen by validation accuracy after dynamic-rank restoration; "final" denotes the terminal allocation trajectory before that checkpoint is reloaded. We pair accuracy only with selected-checkpoint counts. This is a matched local reproduction, not a direct numerical comparison with differently configured results in the original IncreLoRA paper.

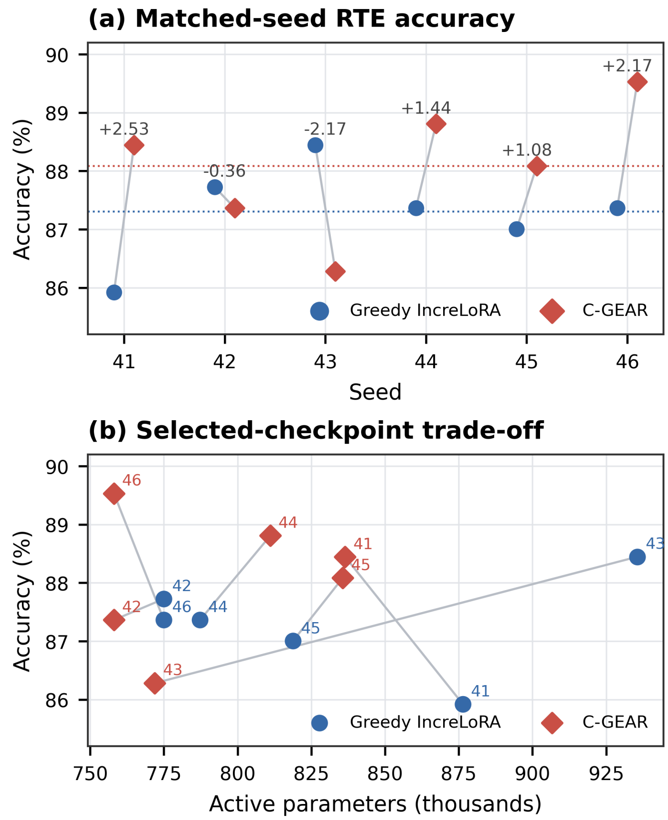


**Figure 2.** Matched results. (a) Per-seed accuracy and C-GEAR minus Greedy differences in percentage points. (b) Accuracy versus active parameters at the selected checkpoint; each gray segment joins a matched seed.

## 5 Results & Contributions

Table 1 gives the central comparison. C-GEAR raises mean accuracy by 0.78 percentage points and wins/ties/loses 4/0/2 matched seeds. At those *same selected checkpoints*, mean active parameters fall from 828,005 to 795,225 and mean active rank from 99.0 to 86.7. The primary paired reduction is 3.55%; the reduction computed from aggregate mean counts is 3.96%, illustrating why the paired statistic must be labeled.

Figure 2 exposes rather than hides variation: C-GEAR loses on seeds 42 and 43, and selected efficiency is not lower for every pair. Nonetheless, the mean selected-checkpoint trade-off favors C-GEAR. Separately, final trajectories average 804,060 active parameters and rank 89.7, versus Greedy's 933,650 and rank 144; the mean paired final-parameter reduction is 13.86%. These terminal counts do *not* inherit selected-checkpoint accuracy.

Telemetry supports the mechanism-level interpretation (Fig. 3). Greedy reaches rank 144 in every run. C-GEAR finishes between ranks 72 and 107, mixes positive event sizes with $k = 0$, and sometimes stops growth at step 300 while the fixed architecture continues optimizing and yields a later best checkpoint. Thus C-GEAR does not merely choose a smaller common target: it learns distinct non-uniform growth schedules. The terminal family profile (Fig. 4) likewise shows redistribution rather than uniform scaling. Calibration and search add a measured 13.4% mean wall-time overhead in this local setup.

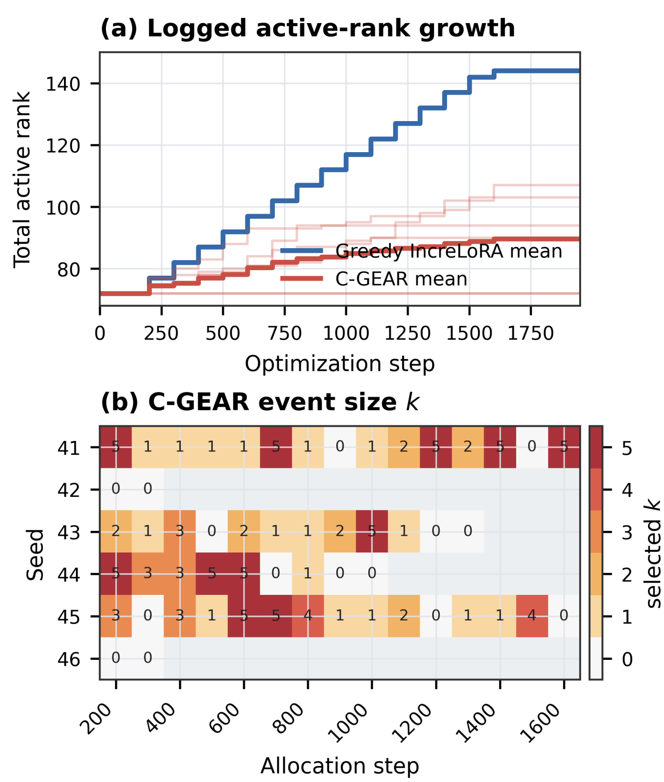


**Figure 3.** Telemetry-derived growth. Top: all six trajectories and method means. Bottom: C-GEAR's selected event size; blank cells occur after allocation stops.

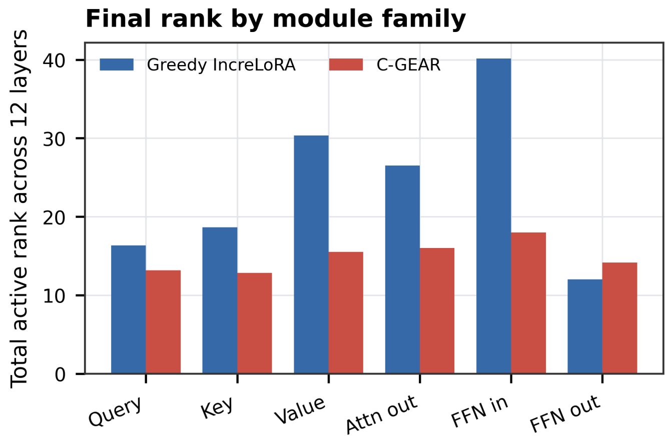


**Figure 4.** Mean terminal rank summed over 12 layers. C-GEAR does not shrink every family proportionally: compared with Greedy, it preserves relatively more FFN-output rank while reducing other families.

## 6 Future Work & Conclusion

The evidence is deliberately narrow: one GLUE task, one backbone, one laptop GPU, and six seeds. It does not establish statistical significance or cross-task generalization. Broader GLUE and model evaluation, larger seed sets, calibration/search ablations, and hardware-aware latency or memory objectives are natural next steps. Within this controlled study, C-GEAR supplies both a methodological improvement and a substantial implementation: it converts compulsory local Greedy growth into calibrated, budget-aware evolutionary decisions. Across six matched RTE seeds, it improves mean accuracy while using fewer selected-checkpoint parameters, and its final architectures grow substantially less.

## References


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